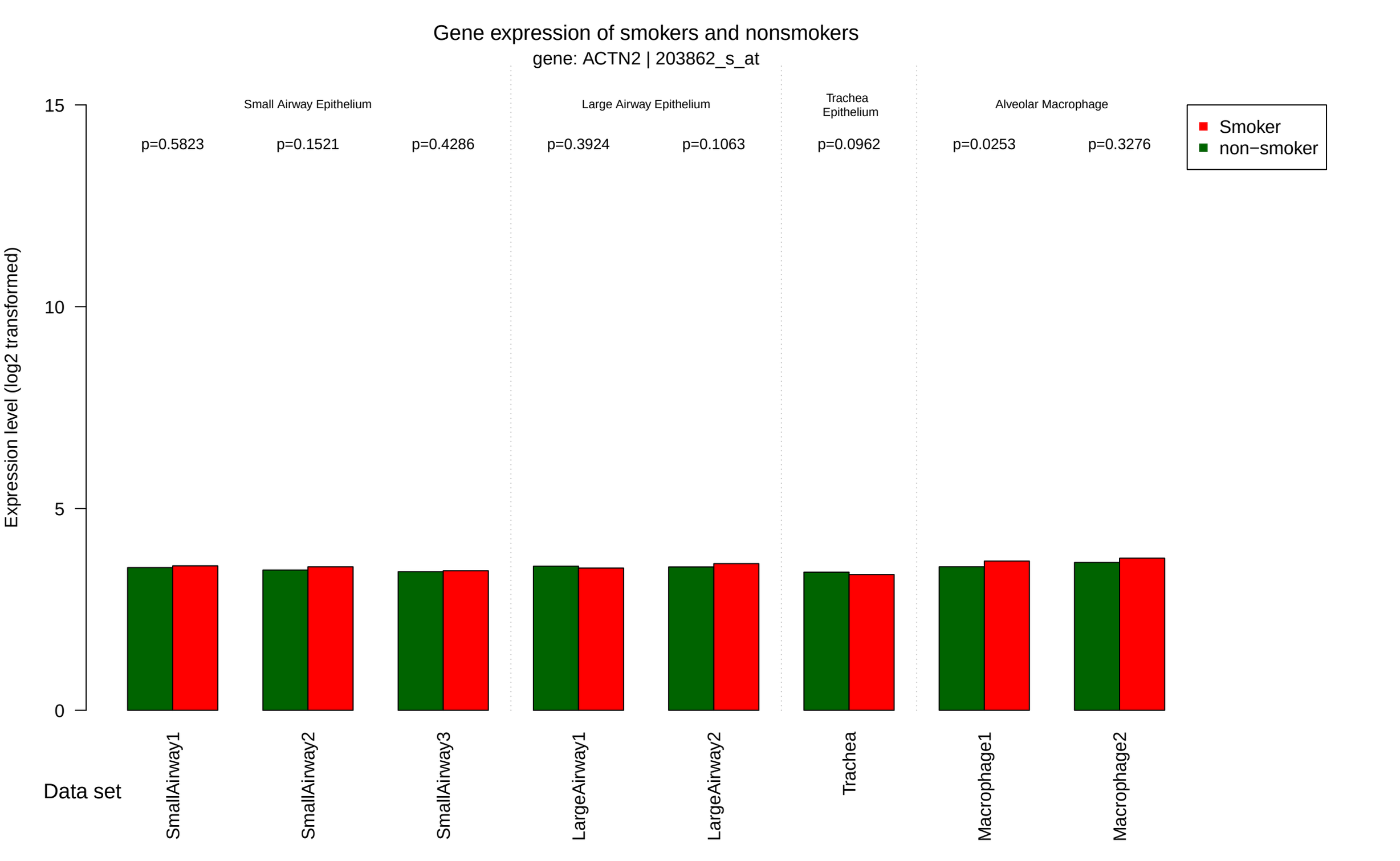
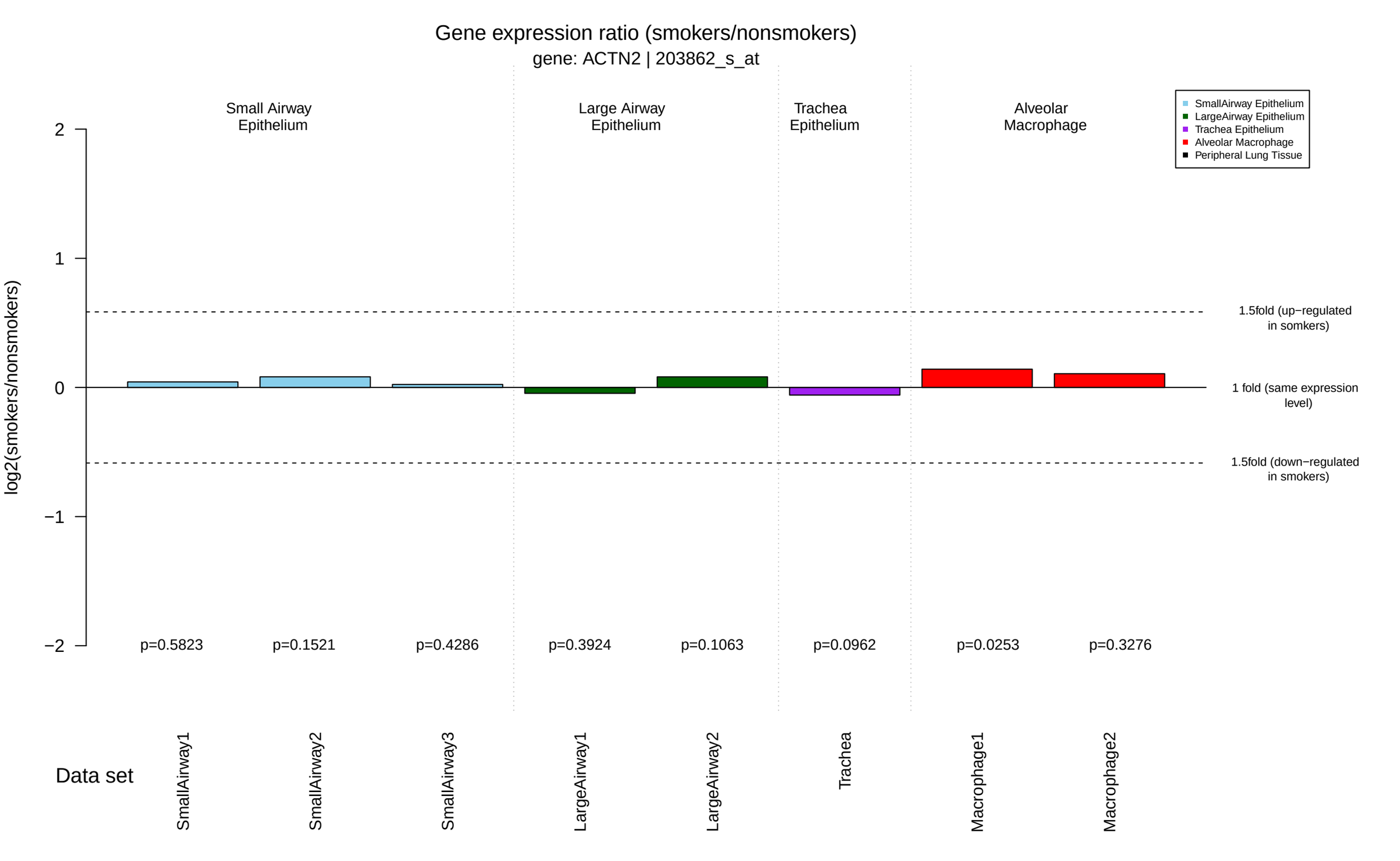
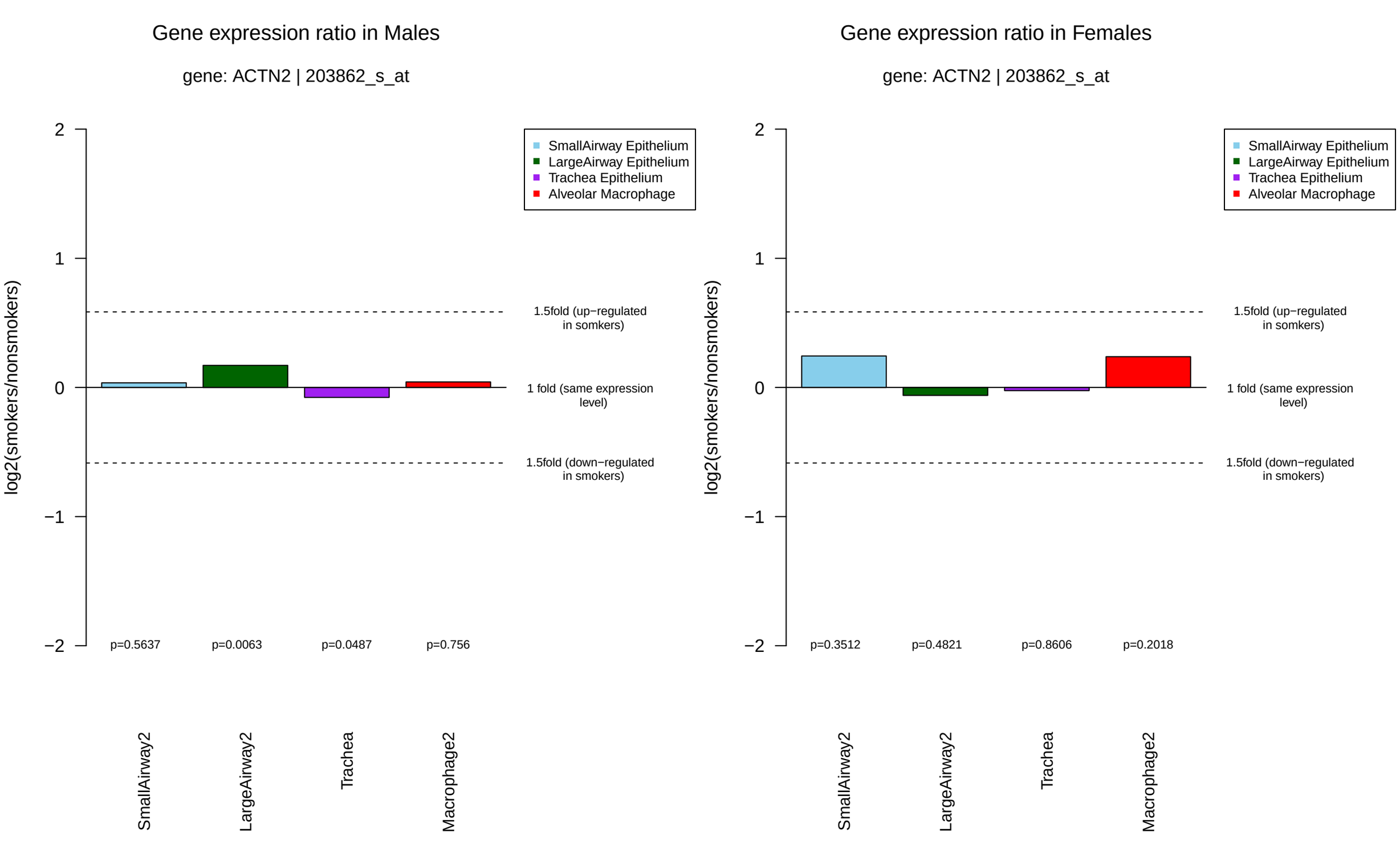
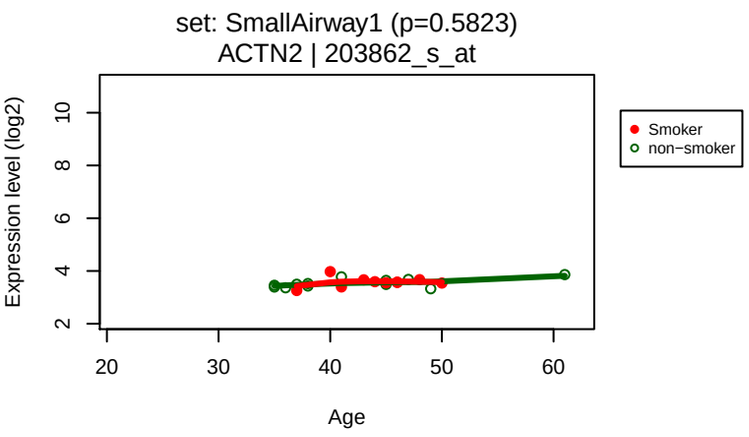
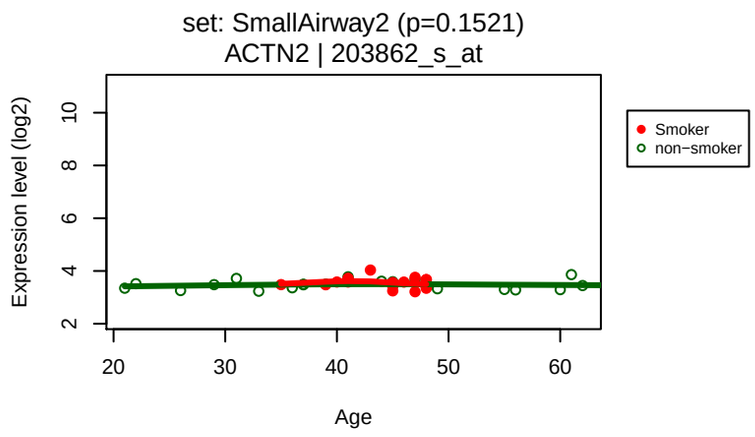
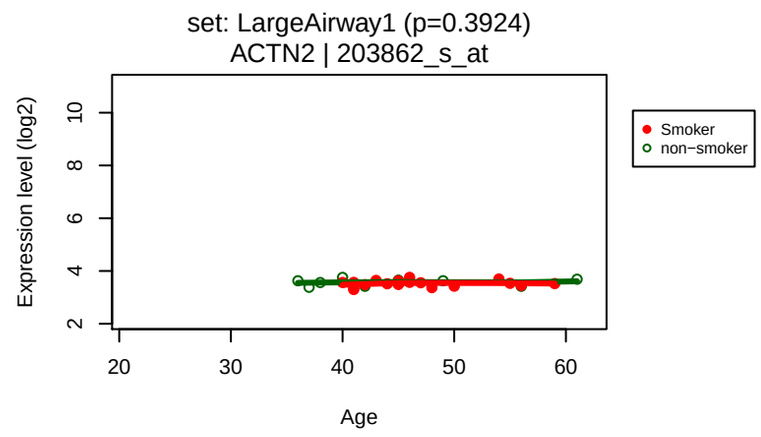

Gene expression of smokers and nonsmokers
gene: ACTN2 | 203862_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5823
p=0.1521
p=0.4286
p=0.3924
p=0.1063
p=0.0962
p=0.0253
p=0.3276
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

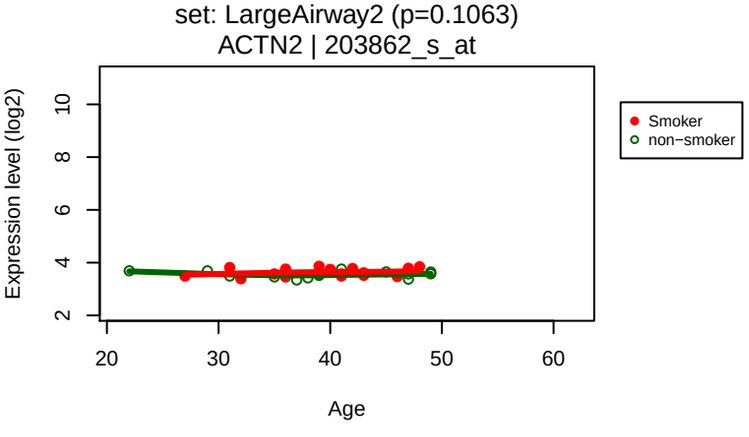

Gene expression ratio (smokers/nonsmokers)
gene: ACTN2 | 203862_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5823
p=0.1521
p=0.4286
p=0.3924
p=0.1063
p=0.0962
p=0.0253
p=0.3276
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

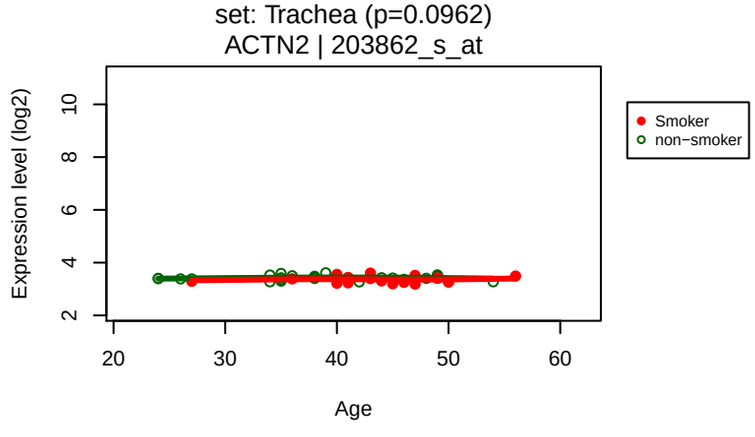

Gene expression ratio in Males
gene: ACTN2 | 203862_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5637
p=0.0063
p=0.0487
p=0.756
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN2 | 203862_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3512
p=0.4821
p=0.8606
p=0.2018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

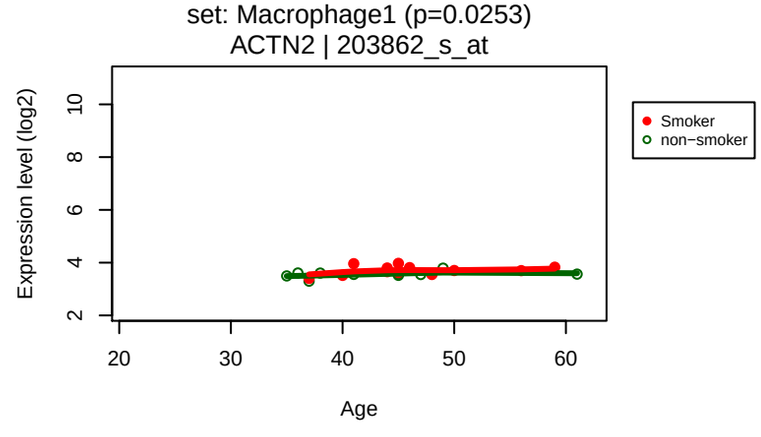

set: SmallAirway1 (p=0.5823)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker

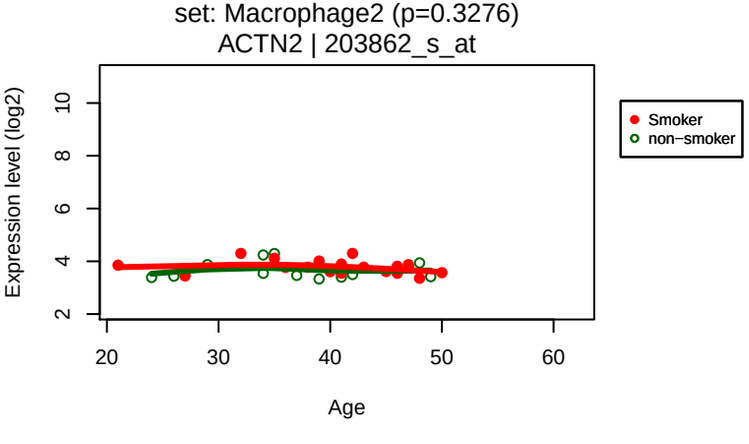

set: SmallAirway2 (p=0.1521)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker

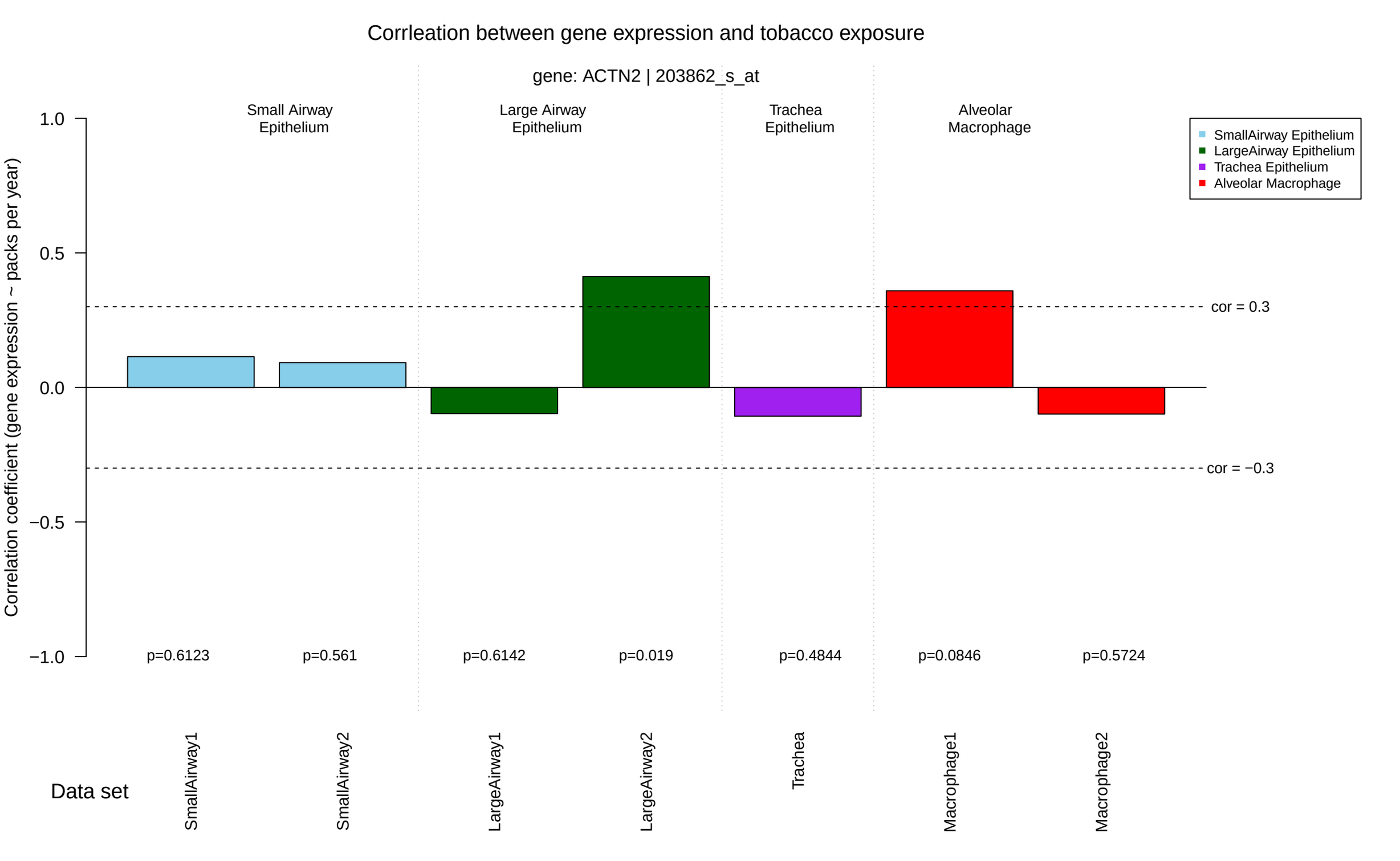

set: LargeAirway1 (p=0.3924)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1063)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0962)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0253)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3276)
ACTN2 | 203862_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN2 | 203862_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6123
p=0.561
p=0.6142
p=0.019
p=0.4844
p=0.0846
p=0.5724
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2